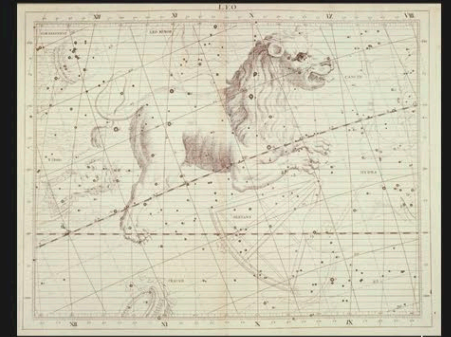


STAR CHARTS AND ATLASES

The first printed star chart was produced in 1515 by the great German artist Albrecht Dürer. Like a celestial globe, Dürer's chart depicted the constellations in reverse, showing the sky as it would be seen from an imaginary position outside the celestial sphere, but before long, charts were being made that could be compared directly with the sky. The finest early star atlas was *Uranometria*, produced in 1603 by the German astronomer

Johann Bayer. This atlas remains one of the most beautiful examples of the celestial cartographer's art. Shortly after its publication,

astronomy was revolutionized by the invention of the telescope. The first major star catalog and atlas of this new era was produced by England's first Astronomer Royal, John Flamsteed (1646–1719). *Atlas Coelestis* shows the Ptolemaic constellations visible from Greenwich, England, based on Flamsteed's own painstaking observations. The pinnacle of celestial mapping came in 1801, when Johann Bode, a German astronomer, published an atlas called *Uranographia*. Covering the entire sky, this atlas depicted over 100 constellations, some invented by Bode himself. Finally, in 1922, a list of 88 constellations was agreed upon by the International Astronomical Union, astronomy's governing body, which also defined the boundaries of each constellation. On modern star charts, the only sign of the traditional pictorial charts are the few lines that link the main stars, suggesting the overall shape of each constellation.



SKETCHY FIGURES

Leo, the Lion, an easily recognizable constellation of the zodiac, is here depicted on the *Atlas Coelestis*, by English astronomer John Flamsteed, published in 1729.



PTOLEMY

Ptolemy (c. 100–170 CE) lived and worked in the great metropolis of Alexandria, Egypt, which was then part of the Greek empire. He was one of the last—and the greatest—of the ancient Greek astronomers. His Earth-centered model of the universe, outlined in the treatise *Almagest*, dominated astronomical theory for 1,400 years. Ptolemy also made a catalog of 1,022 stars in 48 constellations, based on earlier work by Hipparchus.



HEAVENLY PICTURE BOOK

Ancient people imagined gods, heroes, and beasts among the stars, and these figures were depicted on star charts until the 19th century. These charts, from John Flamsteed's *Atlas Coelestis* (1729), show those 48 constellations known to the ancient Greeks depicted on the northern and southern halves of the sky.